

SESSION – 9

Group Anagrams

```
class Solution {
    public List<List<String>> groupAnagrams(String[] strs) {

        HashMap<String, List<String>> map = new HashMap<>();

        for (String str : strs) {

            char[] ch = str.toCharArray();
            Arrays.sort(ch);

            String key = new String(ch);

            map.putIfAbsent(key, new ArrayList<>());
            map.get(key).add(str);
        }

        return new ArrayList<>(map.values());
    }
}
```

The screenshot shows a LeetCode submission interface for the problem "49. Group Anagrams". The problem description states: "Given an array of strings `strs`, group the **anagrams** together. You can return the answer in **any order**." An example is provided: Input: `strs = ["eat","tea","tan","ate","nat","bat"]`, Output: `[["bat"],["nat","tan"],["ate","eat","tea"]]`. The explanation notes that "bat" is unique, while "eat", "tea", and "tan" are anagrams of each other, and "ate", "eat", and "tea" are anagrams of each other. The submission status is "Accepted" with a runtime of 0 ms. The test result shows three cases passed. The code editor displays the following Java code:

```
1 class Solution {
2     public List<List<String>> groupAnagrams(String[] strs) {
3
4         HashMap<String, List<String>> map = new HashMap<>();
5
6         for (String str : strs) {
7
8             char[] ch = str.toCharArray();
9             Arrays.sort(ch);
10
11             String key = new String(ch);
12
13             map.putIfAbsent(key, new ArrayList<>());
14             map.get(key).add(str);
15         }
16
17         return new ArrayList<>(map.values());
18     }
19 }
```